

## Totally Unimodular Matrix (TUM)

• A <sup>n</sup>×<sup>n</sup> square matrix  $A$  with entries  $A_{ij} \in \{-1, 0, 1\}$  is called Totally Unimodular, if  $(A')_{ij} \in \{-1, 0, 1\}$

Def'n:  $A$  is TU if every square submatrix has determinant  $\pm 1, 0, -1$ .  
( $A_{ij} \in \{-1, 0, 1\}$ ).

NB:  $A$  need not be square.

## TUM's Define Integral Polyhedra

Thm: If  $A$  is TU,  $(m \times n)$ ,

and  $b \in \mathbb{Z}^m$ , then

$$P = \{x \mid Ax \leq b\} \subseteq \mathbb{R}^n \text{ has}$$

integral extreme points.

Recall Cramer: If  $M$  is a square  $n \times n$  invertible matrix,

$$Mx = b \rightarrow x = M^{-1}b$$

Also:

$$(M^{-1})_{ij} = \frac{\det(A_{j|i})}{\det A}$$

$x_i = \frac{\det(M_i | b)}{\det M}$ ,  $M_i | b$  is the matrix obtained by replacing  $i^{\text{th}}$  column of  $M$  by  $b$ .

$(n-1) \times (n-1)$

## Proof for TUM & Integrality

$P = \{Ax \leq b\}$  has integral  
ext pts.

An ext pt is defined by  $n$  tight  
ineq. Call these " $B$ "  $B \subseteq [m]$ ,  $|B| = n$ .

$x: A_B^{-1} b_B$  where  $A$  is TU,  $b \in \mathbb{Z}^m$ .

$$x_i = \frac{\det(A_{B_i}, b)}{\det A_B} = \pm \det(A_{B_i}, b) \in \mathbb{Z}$$

$\nwarrow \pm 1$

Equiv:  $x = A_B^{-1} b_B$

$$(A^{-1})_{ij} = \frac{\det(A_{ji})}{\det A}$$



## TUM and Duality

If  $A$  is TU,

$b, c$  are integral vectors of appropriate dimn.

$$\min: c^T x$$

$$\text{st. } \underline{Ax = b}$$

$$\underline{x \geq 0}$$

$$\max: p^T b$$

$$\text{st: } A^T p \leq c$$

both have integral opt. solutions.

## Key Lemma

Lemma : T.F.A.E.

1.  $A$  is TU

2.  $A^T$  is TU

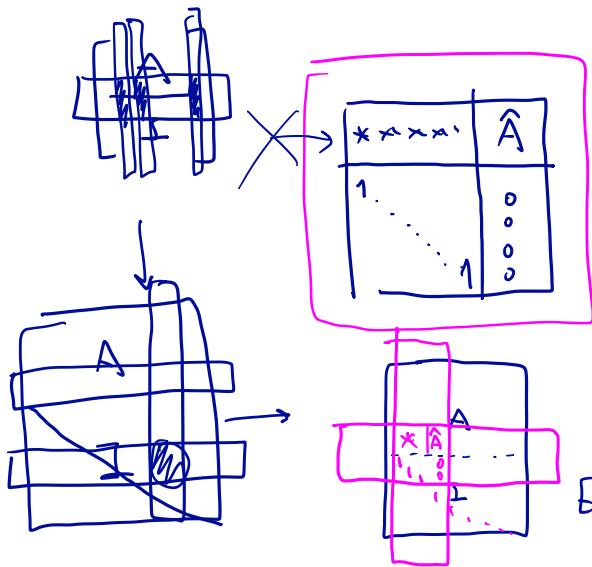
3.  $\begin{bmatrix} A \\ I \end{bmatrix}$  TU

4.  $\begin{bmatrix} I \\ A \\ -A \end{bmatrix}$  TU

Now pf of result  
on prev. slide follows from  
1<sup>st</sup> Thm

# Proof of Key Lemma

$$A \text{ TU} \Rightarrow \begin{bmatrix} A \\ I \end{bmatrix}$$



By exp. by minors,  
and b/c  $\hat{A}$  is a sq.  
submatrix of  $A$ ,  
the det is  $\pm 0, +1, -1$ .

Ex: Fill in the details.

## Integrality implies TUM — 1/2

Thm: If  $A$  has  $m$  rows, then

$$P = \{x : Ax = b, x \geq 0\} \text{ is}$$

integral  $\forall b$  integral iff  $A$

is TU.

$$A = \begin{array}{|c|c|c|} \hline & A_B & \\ \hline \end{array}$$

Enough to show

$A_B$  is TU.

$$\underline{A_B} \text{ is TU} \Leftrightarrow \underline{A_B^{-1}} \text{ is integral} \Leftrightarrow \underline{A_B^{-1}v} \text{ integral for all integral } v.$$

## Integrality implies TUM — 2/2

Need:  $\tilde{A}_B^T v$  is integral when  $v$  is integral.

Choose integral vector  $y$  st.  $\boxed{z = y + \tilde{A}_B^T v \geq 0}$

$$A_B z = \underbrace{A_B y + v}_{\geq 0} \Rightarrow A_B z \text{ is integral}$$

$$A = \begin{bmatrix} A_B & \vdots & A_{B^c} \end{bmatrix}$$

$$\tilde{z} = \begin{bmatrix} z \\ 0 \end{bmatrix}$$

$$A \tilde{z} = \begin{bmatrix} A_B & A_{B^c} \end{bmatrix} \begin{bmatrix} z \\ 0 \end{bmatrix} = A_B z + A_{B^c} 0 = A_B z$$

$$= (A_B y + v) \quad \text{Q.E.D.}$$

Let  $b = A_B z$ , integral

$P = \{Ax = b, x \geq 0\}$ .  $\tilde{z}$  is an ext point of  $P \Rightarrow \tilde{z}$  is integral

$$\tilde{z} = \begin{pmatrix} z \\ 0 \end{pmatrix}, \Rightarrow z \text{ is integral} \Rightarrow \tilde{A}_B^T v \text{ integral} \quad \square$$



## Hoffman and Kruskal (Corollary)

Thm:  $A$  integral  $\{-1, 0, 1\}$

$A \geq TU$  iff  $\forall b$  integral

$P = \{x : Ax \leq b, x \geq 0\}$  is integral.

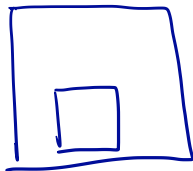
pf. follows from above, and fact that

$A \geq TU \iff [A \ I] \geq TU.$

$$A \in \{1, 0, -1\}$$

$$A^{-1} \in \{1, 0, -1\}$$

$$\det A \in \{1, -1\}$$



$$\det(M_1 M_2) = \det M_1 \cdot \det M_2$$

$$\det A = \frac{1}{\det A^{-1}}$$

$$(A^{-1})_{ij} =$$

$A$   $n \times n$  inv, TU

$$\Rightarrow (A^{-1})_{ij} \in \{0, 1, -1\}$$

$$\det(A^{-1})_{ij} \xrightarrow{\text{sym subm}} \det \{0, 1, -1\}$$

$$\det A \neq \pm 1$$